

Types of Solution and Properties of Solution

Subject / Theme :

Date: / /

Sat ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri

→ Dilute Solution, Solution, types of Solution, units of concentration, mole fraction.

→ Colligative Properties.

→ Raoult's law.

→ Alternative expression of Raoult's Law.

→ Molecular weight from Raoult's law.

→ Ideal and Real solution.

→ Deviation.

→ Henry's law and its application.

→ Nearst Distribution Law.

☐ **Solution:**

A solution is usually defined as a homogeneous mixture of two or more substances whose composition can be varied over certain ranges.

☐ **Types of Solution:**

Solution = Solute + Solvent

1. Gas in gas

2. Gas in liquid

3. Gas in Solid

4. Liquid in liquid

5. Liquid in solid

6. Solid in Solid.

7. Solid in liquid.

Dilute Solution:-

In a solution, if the amount of solute is small and the amount of solvent is large, it is called dilute solution.

Concentrate Solution:-

In a solution, if the amount of solute and solvent both are equal, then it is called concentrate solution.

Unit of Concentration:-

Molarity (mol L^{-1}):-

The molarity of solution is the number of moles of the solute dissolved in 1000 ml of the solution.

If 60g (1 mole) of urea is dissolved in 1000 ml H_2O solution then molarity is 1 molar.

Molality

Molality:-

The molality of solution is the number of moles of the solute dissolved in 1000g of the solvent.

60g urea (1 mole) is dissolved in 1000g H_2O creates 1 molal solution.

Normality:-

The number of gram equivalent to solute that is present in a one-liter solution.

☐ Mole fraction:-

Mole fraction represents the number of molecules of a particular component in a mixture divided by the total number of moles in the given mixture.

$$X_A = \frac{n_A}{n_A + n_B}$$

☐ Liquid to Liquid Solution:-

When two liquids are mixed together there is an energy change involved in the system. This is due to the movement of the solvents and solute molecules to form solutions.

Sometimes energy is absorbed and sometimes energy is released.

☐ Colligative Solution:-

The word colligative ~~solution~~ is derived from latin word colligatus that means to bind together or going together.

The essential part of this is that they depend only on the number of solute particle present in solution.

Defⁿ: Those properties of solutions which depends only upon the total number of molecules of solute per unit volume and on its chemical nature and being closely

related to each other through a common explanation are called colligative properties.

Four main colligative properties of solutions are follows:-

1. Lowering of vapour pressure.

2. Elevation of boiling point

3. Depression of freezing point.

4. Osmotic pressure.

The colligative properties are widely used for the determination of molecular mass of substances.

Elevation of boiling point is a colligative property:

The thermodynamic expression of elevation of boiling point given as below:

$$\Delta T_b = \frac{RT_b^2}{L_e} \times X_2$$

$$\Delta T_b = \frac{RT_b^2}{L_e} \times \frac{n_2}{n_1} \quad \left[\text{For dilute solution, } X_2 = \frac{n_2}{n_1} \right]$$

From the above equation, it is clear that, boiling points depend only on the number of moles but does not depend on its nature.

Thus, the elevation of boiling point is a colligative property.

Subject / Theme:

Pressure is colligative property

□ The lowering of vapour pressure is colligative property

According to the Raoult's law for lowering of vapour

Pressure:-

"The relative lowering of vapour pressure of a solvent by a non volatile solute is equal to the mole fraction of the solute in solution"

Mathematically it may be written;

$$\frac{p_0 - p}{p_0} = \frac{n_2}{n_1 + n_2}$$

where,

p_0 = vapour pressure of solvent.

p = vapour pressure of solution.

n_1 = mole number of solvent.

n_2 = mole number of solution.

From the above equation, it is clear that the lowering of vapour pressure depends only on the mole number of solute in a solution but independent of its nature. Thus the lowering of vapour pressure is colligative property.

Rault's Law:-

The statement of Rault's Law:-

"The relative lowering of vapour pressure of a solvent due to the addition of a non-volatile and non-electrolyte solute is equal to the mole fraction of the solute present in the solution."

Explanation:-

If p_0 and p are the vapour pressure of the solvent and the solution respectively, the relative lowering of vapour pressure is $\frac{p_0 - p}{p_0}$. In a solution containing n_2 moles of solute in n_1 moles of solvent, the mole fraction of the solution is $\frac{n_2}{n_1 + n_2}$.

Then according to Raoult's Law,

$$\frac{p_0 - p}{p_0} = \frac{n_2}{n_1 + n_2} = x_2$$

Where, x_2 :-

x_2 is the mole of the solute. This is the mathematic expression of Raoult's law.

For very dilute solution, n_2 can be neglected in comparison with n_1 and n_0 .

Raoult's taken the form, $\boxed{\frac{p_0 - p}{p_0} = \frac{n_2}{n_1}}$

Alternative expression for Raoult's law:-

From the mathematical expression of Raoult's law of lowering of vapour pressure, we get,

$$\frac{p_0 - p}{p_0} = x_2$$

$$\Rightarrow \frac{p_0}{p_0} - \frac{p}{p_0} = x_2$$

$$\Rightarrow 1 - \frac{p}{p_0} = x_2$$

$$\Rightarrow \frac{p}{p_0} = 1 - x_2 \quad \text{--- (i)}$$

If x_1 is the mole fraction of solvent and x_2 is that of the single solute, it follows that for a binary solution,

$$x_1 + x_2 = 1$$

$$\Rightarrow x_1 = 1 - x_2$$

Now from eqn (i) we get,

$$\frac{p}{p_0} = 1 \cdot x_1$$

$$\Rightarrow p = p_0 x_1 \quad \text{--- (ii)}$$

$$p \propto x_1$$

Thus, the vapour pressure of solution is directly proportional to the mole fraction of solvent. This is taken as an alternative way of stating Raoult's law.

The vapour pressure of the solvent over any solution is equal to the vapour pressure of the pure solvent multiplied by the mole fraction of the solvent.

☐ Molecular weight from Raoult's law of lowering of vapour pressure:-

From Raoult's Law,

$$\frac{p_0 - p}{p_0} = \frac{n_2}{n_1 + n_2} = X_2$$

For a dilute solution,

$\frac{n_2}{n_1 + n_2}$ may be replaced by $\frac{n_2}{n_1}$. As, n_2 is negligible.

$$\frac{p_0 - p}{p_0} = \frac{n_2}{n_1}$$

If W_2 gm of solute of molecular weight M_2 be present W_1 gm of solvent of molecular weight M_1 then,

$$\frac{n_1}{n_2} = \frac{W_2/M_2}{W_1/M_1}$$

And the equation becomes,

$$\frac{p_0 - p}{p_0} = \frac{W_2/M_2}{W_1/M_1}$$

By meaning p_0 and p for a solution of known

concentration M_2 may be obtained.

Ideal Solution:-

The solution which obey Raoult's law at any temperature and pressure is called ideal solution.

When ideal solution is prepared -

1. It doesn't absorb heat.
2. It doesn't evolve heat.
3. No volume change occurs.

Thermodynamic criteria of ideal solution:-

1. Volume of mixing, $\Delta V_{mix} = 0$
2. Enthalpy of mixing, $\Delta H_{mix} = 0$
3. Entropy of mixing, $\Delta S_{mix} = -R \sum x_i \ln x_i$

When the ideal solution is prepared by two compound A and B. The intermolecular forces between A-A and B-B and A-B must be same.

Non-ideal Solution:-

A solution which shows deviation from Raoult's law is called non-ideal solution.

Reason of solution may not behave ideally:-

1. If the concentration of the solution be high.
2. If the solution is associated or disassociated in solution.
3. If the internal pressure be different in solution.
4. If compound is formed between solvent and solute

It doesn't follow Raoult's law $\Delta V_{mix} \neq 0$ and $\Delta H_{mix} \neq 0$

V.P composition plots of the system:-

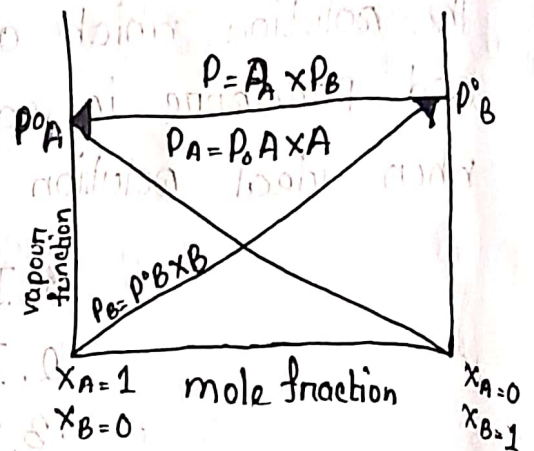
Curve for ideal solution:-

$$\Delta H_{mix} = 0$$

$$\Delta V_{mix} = 0$$

$$\Delta S_{mix} = 0$$

$$A-B = A-A = A=B$$



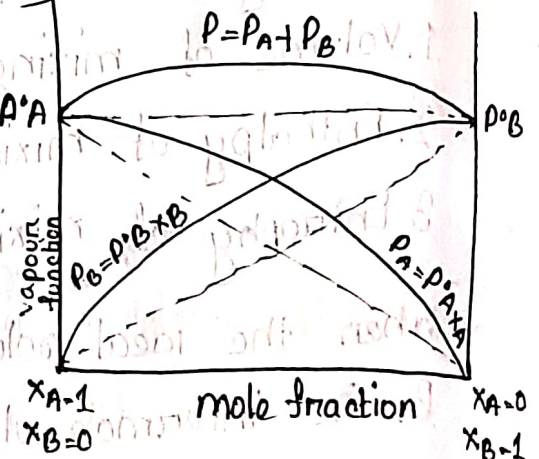
Curve for non-ideal solution:- (Positive)

$$\Delta H_{mix} = +ve$$

$$\Delta V_{mix} = +ve$$

$$\Delta S_{mix} = +ve$$

$$A-B > A-A / B-B$$



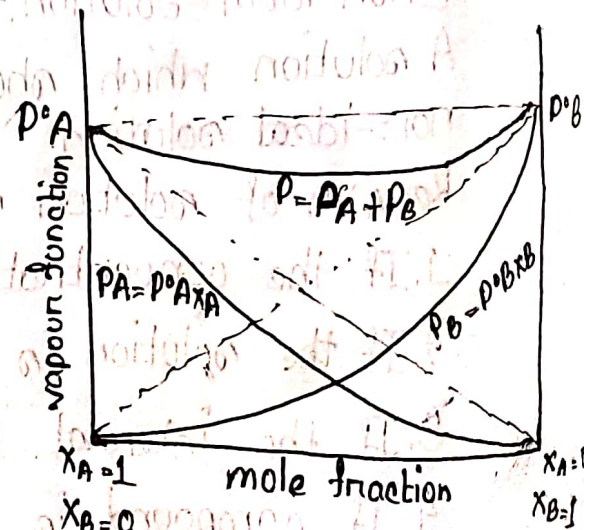
Curve for non-ideal solution:- (Negative):-

$$\Delta H_{mix} = -ve$$

$$\Delta V_{mix} = -ve$$

$$\Delta S_{mix} = -ve$$

$$A-B < A-A / B-B$$



Positive Deviation:-

Positive deviation from Raoult's law means that partial vapour pressure exerted by a component is greater than that exerted by Raoult's law in ideal solution.

$$\text{i.e. } P_A > P_A^\circ \times X_A$$

Criteria:-

1. A-B interactions lower than A-A and B-B interactions.
2. $\Delta H_{\text{mix}} = +ve$ (endothermic)
3. $\Delta V_{\text{mix}} = +ve$

Physical Significance:-

The positive deviation from the ideal behaviour occurs due to repulsion between solvent and solute molecule mutually and thereby the Δu of the molecule increases which causes high vapour pressure.

Ex - Aceton - CS_2 , C_6H_{12} - CCl_4

Subject / Theme :

Negative deviation:-

Negative deviation from Raoult's law means that partial vapour pressure exerted by a component is less than that exerted by the Raoult's law in ideal solution.

i.e. $P_A < P_A^0 \times A$

Criteria:-

1. A-B interaction is greater than A-A and B-B interaction
2. $\Delta H_{mix} = -ve$ (exothermic)
3. $\Delta V_{mix} = -ve$

Physical Significance:-

The reason for negative deviation is mutual attraction between solute and solvent molecules. This causes the evolution of heat and the volume of the mixture decreases and vapour pressure also decreases.

Ex - Acetone - chloroform
water - methanol.

Henry's Law:

The mass of gas dissolved by a given volume of solvent, at constant temperature is proportional to the pressure of the gas which it is in equilibrium.

It stated mathematically,

$$w \propto p$$

$$\Rightarrow w = Kp$$

Where,

w = Mass of the gas dissolved in a unit volume of the solvent.

p = Pressure of the gas in equilibrium with the solvent.

K = constant, called Henry's law constant.

Application:-

The influence of partial pressure on solubility is utilized in making carbonated beverages like beer, champagne and many soft drinks. So called soda water is bottled under a carbon dioxide pressure of about 4 atm. When the bottle is opened to the air, the partial pressure of CO_2 above the solution decreased about 0.01 atm and bubbles out.

Subject / Theme:

Limitation and validity of Henry's law:-

1. It is obeyed fairly satisfactorily by many gases of low solubility, provided the pressure is not too high as to the temperature, or not too low.
2. It holds good for dilute solutions of gases which do not react with solvent.
3. This law is strictly applicable to those gases where the molecular species are the same in the liquid phases.

Or

1. It applies closely to gases with nearly ideal behavior.

1. at moderate temperature and pressure.

2. If the solubility of the gas in the solvent is low,

3. the gas doesn't react with the solvent to form a new species.

4. The gas doesn't associate or dissociate on dissolving in the solvent.

Van't Hoff Isotherm:-

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -2.303RT \log K$$

The equation is known as Van't Hoff Isotherm. It may be used to calculate the change in free energy of a reaction in the standard (ΔG°) from the equilibrium constant and vice versa.

The sign of ΔG° indicates whether the forward or backward reaction is spontaneous. Considering the equation we can have three possibilities depending on the sign of ΔG° for the reaction.

1. If ΔG° is negative, $\log K$ must be positive. The reverse reaction proceeds spontaneously in the forward reaction.

2. If ΔG° is positive, $\log K$ must be negative and K is less than one. The reverse reaction is spontaneous.

3. If $\Delta G^\circ = 0$, $\log K = 0$ and $K = 1$. The reaction is at equilibrium.